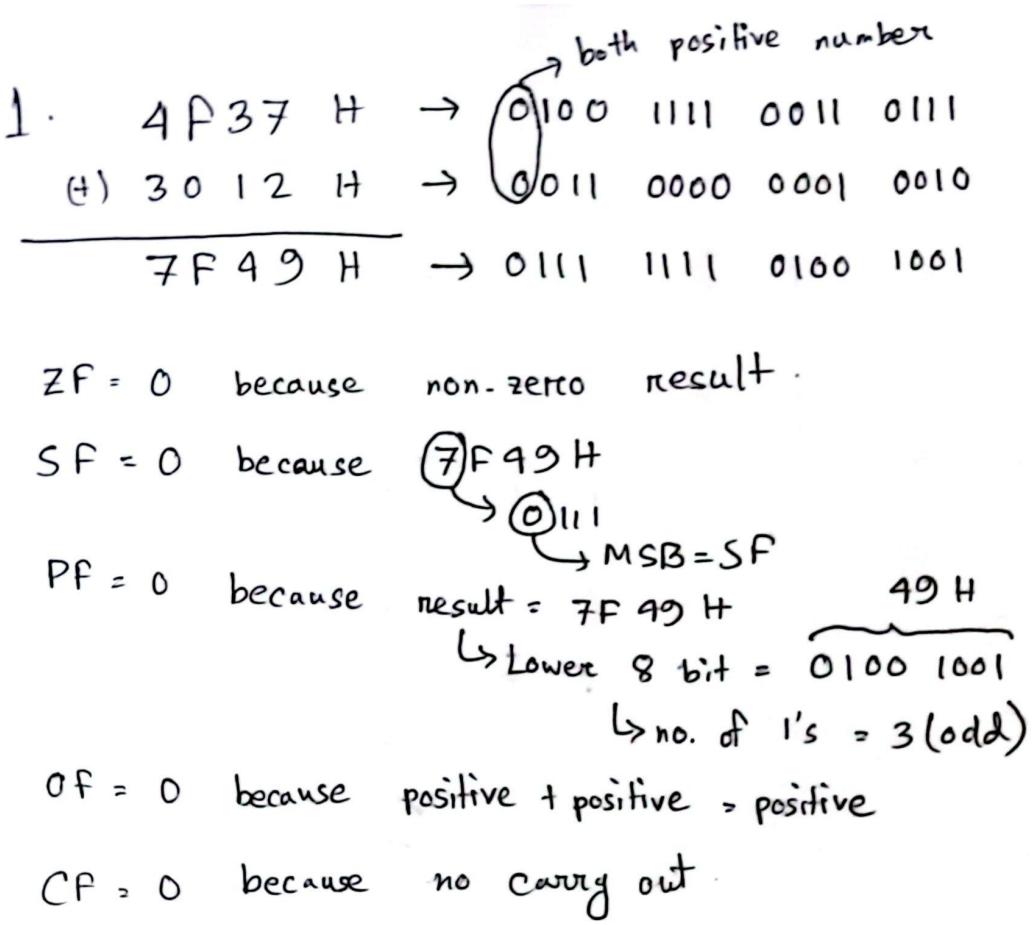
Solution of Assignment - 01

1.



2.

I. for DS, we need BX register

DS : BS

34A9H : 8722H

Physical Address

$$\begin{array}{r} 34A90\ H \\ 8722\ H \\ \hline 3D1B2\ H \end{array}$$

a) Segment Register = 3D1B2H − 7B85H

= 3562DH

after right shift = 3562H ~~(right sh~~

Its not valid because if we add the offset with the left shifted value 35620H we cant get our original physical address

b) Segment Register = 3D1B2H - 45D2H

= 38BE0H

after right shift = 38BEH

This is valid because after left shift and adding the offset to 38BE0H we get our original physical address back.

c) Offset register = 3D1B2H - 7D920H

= FFFBF892H

This is a negative value and offset cant be negative. In addition offset can only be 16 bits.

3.

3. $2^n \times m$ = memory capacity

n = Address bus bit

m = Data bus size

128 MB = 128 × 1 MB

$= (2^7 \times 2^{20})$ bytes

$= 2^{27}$ bytes

For RAM-A,

Data bus = 8 bit = 1 byte

$\therefore 2^n \times 1 = 2^{27} \Rightarrow 2^n = 2^{27}$

$\therefore n = 27$

For RAM-B,

Data bus = 16 bit = 2 byte

$\therefore 2^n \times 2 = 2^{27} \Rightarrow 2^n = 2^{26}$

$\therefore n = 26$

4. a)

4. a) $150 + x + 550 = 13335083$

$\therefore x = 13334383 \text{ ns} \times 10^{-9}$

$= 0.013334383 \text{ s}$

So,

In 0.013334383s data transferred 2048 byte

$\therefore$ " 1 s " " $\frac{2048}{0.013334383}$ bytes

$= 153587.9088$ bytes

$= \frac{153587.9088}{1024}$ KB

$= 149.9881922$ KB

$\simeq 150$ KB

b)

b] At first we need to calculate how many system do we need here,

Data transfer in happening in this manner = 1+2+3+ -- +n

So,

$1+2+3+\dots+n = 7875$

$\Rightarrow \frac{n(n+1)}{2} = 7875$

$\Rightarrow n^2+n-15750 = 0$

$n = 125$ and $n = -126$ [We don't consider negative value here]

So data ~~is train~~ data transfer time $= \left(150 + \frac{1\times10^9}{150\times1024} + 550\right) +$

$\left(150 + \frac{2\times10^9}{150\times1024} + 550\right) +$

$- - - - - - - -$

$\left(150 + \frac{125\times10^9}{150\times1024} + 550\right)\}$

$= 125\times700 + \frac{10^9}{150\times1024}(1+2+\dots+125)$

$= 125\times700 + \frac{10^9}{150\times1024}\times\frac{125\times126}{2}$

$= 0.0513$ s Ans:

5.

5. 8BCD H → (1) 000 10 11 1100 1101 — (1) → ~~positive~~ Negative number

(+) 71AB H → (0) 111 0001 1010 1011 — (0) → positive number

1A22 H → 0001 1010 0010 0010

CF = 0 because no carry out.

ZF = 0 because non-zero result.

SF = 0 because (1)A22 H → (0)001 → MSB = SF

PF = 1 because result = 1A22 H → Lower 8 bit = 0010 0010 (22H) → no. of 1's = 2 (even)

OF = 1 because (negative − positive) should be negative but result is positive

6.

6. 68E2H : 3B9AH

Physical address - 68E20H + 3B9AH
= 6C9BAH

a) Segment Register: 6C9BAH - 4C5AH
= 67060H

After rightshift = 6706H

This is valid as after leftshift we can add our offset to get back the physical address.

b) Segment register = 6C9BAH - 7D98H
= 64C22H

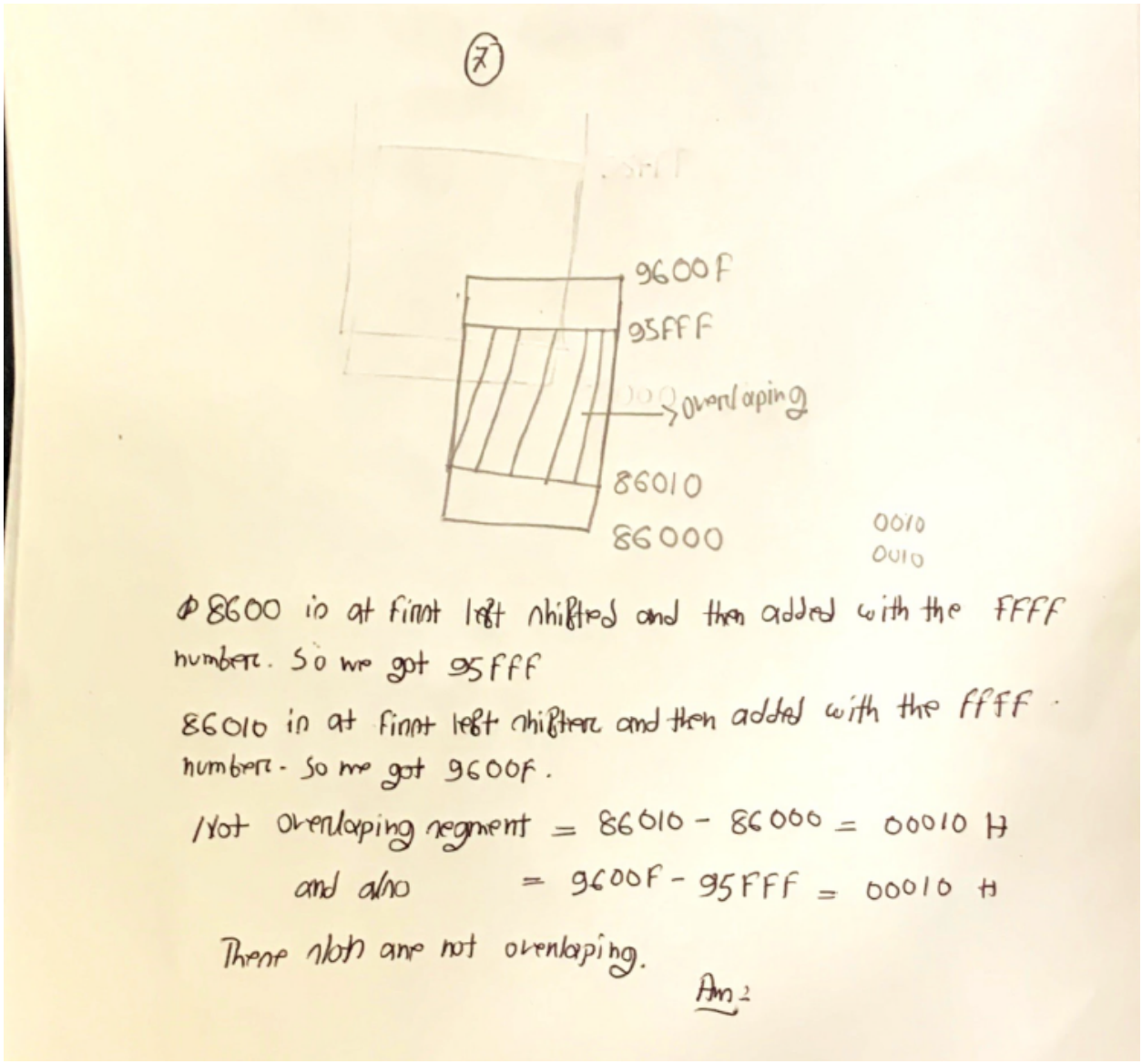
after rightshift = 64C2H

No valid as after left shift 64C20H, we cannot get back physical address after adding offset

c) Offset register - 6C9BAH - 26FF0H
= 459CA

This is not a valid offset, since it is 20 bit offset register should be only 16 bit.

7.



0010 H = 16 (in decimal)
So, number of non overlapping physical address slots are 16+16 = 32